MAKING THE GENERIC FIBRE GEOMETRICALLY REDUCED BY A FINITE UNIVERSAL HOMEOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6718-6740 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 6742-6769, the complete author proof (26 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Reductions of schemes, finite universal homeomorphisms, flatness and finite presentation, and generic fibres; the construction is retained.

Source reference. Original lemma 0BRA: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6769–6769. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-reduction-generic-fibre: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6690–6694. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 054M: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6769–6769. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 052A: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6769–6769. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04KT: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6769–6769. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Assume that Y is irreducible and f is of finite type. There exists a diagram*

$$\begin{array}{ccccc} X' & \xrightarrow{g'} & X_V & \longrightarrow & X \\ {\scriptstyle f'}\downarrow & & \downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & V & \longrightarrow & Y \end{array}$$

where

(1) *V is a nonempty open of Y,*
(2) *$X_V = V \times_Y X$,*
(3) *$g : Y' \to V$ is a finite universal homeomorphism,*
(4) *$X' = (Y' \times_Y X)_{red} = (Y' \times_V X_V)_{red}$,*
(5) *g' is a finite universal homeomorphism,*
(6) *Y' is an integral affine scheme,*
(7) *f' is flat and of finite presentation, and*
(8) *the generic fibre of f' is geometrically reduced.*

Proof. Let $V = \mathrm{Spec}(A)$ be a nonempty affine open of Y. By assumption the nilradical $\mathfrak{p} = \sqrt{(0)}$ of A is a prime ideal. Let $K = \kappa(\mathfrak{p})$. Let p be the characteristic of K if positive and 1 if the characteristic is zero. By Varieties, Lemma 04KT there exists a finite purely inseparable field extension K'/K such that $(X_{K'})_{red}$ is geometrically reduced over K'. Choose elements $x_1, \ldots, x_n \in K'$ which generate K' over K and such that some p-power of x_i is in $A/\mathfrak{p}$. Let $A' \subset K'$ be the finite A-subalgebra of K' generated by $x_1, \ldots, x_n$. Note that A' is a domain with fraction field K'. By Algebra, Lemma 0BRA we see that $A \to A'$ induces a universal homeomorphism on spectra. Set $Y' = \mathrm{Spec}(A')$. Set $X' = (Y' \times_Y X)_{red}$. The generic fibre of $X' \to Y'$ is $(X_{K'})_{red}$ by Lemma lemma-reduction-generic-fibre which is geometrically reduced by construction. Note that $X' \to X_V$ is a finite universal homeomorphism as the composition of the reduction morphism $X' \to Y' \times_Y X$ (see Morphisms, Lemma 054M) and the base change of g. At this point all of the properties of the lemma hold except for possibly (7). This can be achieved by shrinking Y' and hence V, see Morphisms, Proposition 052A. □